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Electropolishing of Bi_2Te_3 based alloys

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Abstract

Polarography has been used as a tool for controlling the electropolishing of p-type $Bi_{0.5}Sb_{1.5}Te_{3.0}$ and n-type $Bi_2Se_{0.3}Te_{2.7}$: 0.2 wt.% SbI_3 semiconducting materials used in thermoelectric coolers at particular current density in terms of electropolished film thickness preceded by the electroetching of these materials in suitable electrolytes, by measuring the diffusion current for the reduction of Sb and Se ions at dropping mercury electrode (DME) from electropolishing electrolytes for the p-type and n-type materials respectively. In addition, *I–V* measurements for correlating the current density with the growth of the electropolished film for these n-type and p-type semiconductors have been carried out. © 2001 Elsevier Science B.V. All rights reserved.

Keywords: Dropping mercury electrode; Bi_2Te_3 based alloys; Electropolishing

1. Introduction

Bi_2Te_3 based alloys are commonly used for fabrication of thermoelectric coolers near room temperature applications [1]. The temperature differential, ΔT, achieved using these devices besides depending upon thermoelectric figure of merit Z of p-type and n-type materials also depends to a great extent on the contact resistance between thermoelements and metal pads. Electropolishing is a prerequisite step to remove subsurface damage caused due to polishing with alumina of successively lower grades thereby reducing the contact resistance between the thermoelements and the metal pads. This process renders the semiconductor surface atomically clean free from surface states, impurities etc.

In this paper, we report results of electropolishing on both p- and n-type samples of composition $Bi_{0.5}Sb_{1.5}Te_{3.0}$ and $Bi_2Se_{0.3}Te_{2.7}$ doped with 0.2 wt.% SbI_3, respectively. *I–V* measurements during electropolishing of the samples have been carried out to optimize current density for both p and n-type samples. Surface of the electropolished samples has been examined using SEM. Results of EDAX analysis of electropolished samples have been discussed. Procedure has been developed for accurate etch rate control of electropolished samples using polarographic technique.

2. Experimental procedure

Starting elements Bi, Sb, Te and Se of purity 99.999% were weighed in the appropriate proportions and loaded in quartz ampoule of I.D. 17 mm and sealed under vacuum of $\sim 10^{-6}$ Torr. The sealed ampoule was placed in rocking furnace. The temperature of the furnace was slowly raised to 650°C. The furnace was kept at this temperature for 15 h. The furnace was thereafter rocked for 4 h to ensure thorough mixing of the melt. The ampoule was then lowered at growth rate of 2 mm/h and under a temperature gradient of ~25°C near solid–liquid interface using directional freezing technique. Electropolishing solution used for p-type samples contained 90 g KOH, 56 g tartaric acid, 220 ml glycerol and rest DI water to make 1 l of solution. Electropolishing solution used for n-type samples contained 83 g NaOH, 67 g tartaric acid, 100 ml glycerol and rest DI water to make 1 l of solution. High purity graphite was used as the cathode and sample surface used as the anode in electropolishing experiments. The samples were thoroughly washed with DI water after electropolishing experiments before using them for further evaluation.

3. Results and discussion

Smooth electropolished surfaces were observed, the smoothness of the surfaces being intercepted at the grain

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boundaries by the appearance of pits. It may be mentioned that Ellis [2] has also reported pit formation at the grain boundaries while performing experiments of electrolytic etching in germanium. EDAX analysis of Bi, Sb and Te in p-type $Bi_{0.5}Sb_{1.5}Te_{3.0}$ and Bi, Se and Te in n-type $Bi_2Se_{0.3}Te_{2.7}$: 0.2 wt. SbI_3 does not reveal preferential dissolution of any of the constituent elements in the solution. It may be mentioned that chemical etching with bromine–methanol solution in mercury cadmium telluride leaves the sample surface tellurium rich while at the same time on the cation sites it leaves the surface mercury rich. However, no observation of this kind was observed in both p-type as well as n-type samples.

The *I–V* curve in Fig. 1 shows anodic etching, electropolishing followed by oxygen evolution in the later part of the curve of n-type $Bi_2Se_{0.3}Te_{2.7}$: 0.2 wt.% SbI_3. The initial rapid rise in current density (part AB) indicates anodic dissolution of selenium possibly as SeO_2 which goes in to solution showing its acidic nature as H_2SeO_3. Fig. 2 shows polarographic analysis of selenium content in the electrolyte left after electropolishing. The process of electropolishing in the curve (parts CD and EF in Fig. 1) follows anodic etching. The anodic current density has been found to be less varying in the region of electropolishing. The constancy in current density in two different regions indicates the mechanism of anodic oxide formation of two distinct components i.e. bismuth and tellurium in the case of n-type material. In this way, the behaviour of the current density in the electropolishing region w.r.t. applied voltage in the case of electropolishing of metals [3] is self-explanatory because of the anodic polishing by way of oxide formation of the only component (metal) subjected to positive electric field, using particular electrolyte during the course of electropolishing.

Fig. 3 shows the *I–V* curve of electropolishing of p-type $Bi_{0.5}Sb_{1.5}Te_{3.0}$. The initial rapid rise in current density is less pronounced because of the absence of selenium in the p-type material. The anodic dissolution of tellurium in the form of etching is manifested by the initial part of the curve AB. Parts CD and EF indicate electropolishing current density ranges. Oxygen evolution follows afterwards in the last region of this curve where current density shoots up again.

Use of polarographic technique was adopted for knowing the quantity of the etched material in the electropolished electrolyte. Figs. 4 and 5 indicate polarographic analysis curves for detecting antimony and tellurium respectively in the electropolished electrolyte. Amount of antimony detected in the electropolished electrolyte while carrying out electropolishing of p-type $Bi_{0.5}Sb_{1.5}Te_{3.0}$ gives the quantitative control related to the control of thickness in the electropolished layer during the electropolishing process. In the same way, diffusion current measured by polarograph for knowing the selenium content in the electropolished electrolyte can control the rate of electropolishing in n-type $Bi_2Se_{0.3}Te_{2.7}$: 0.2 wt.% SbI_3.

It may be mentioned that during the electropolishing process glycerine, because of its lower conductivity controls

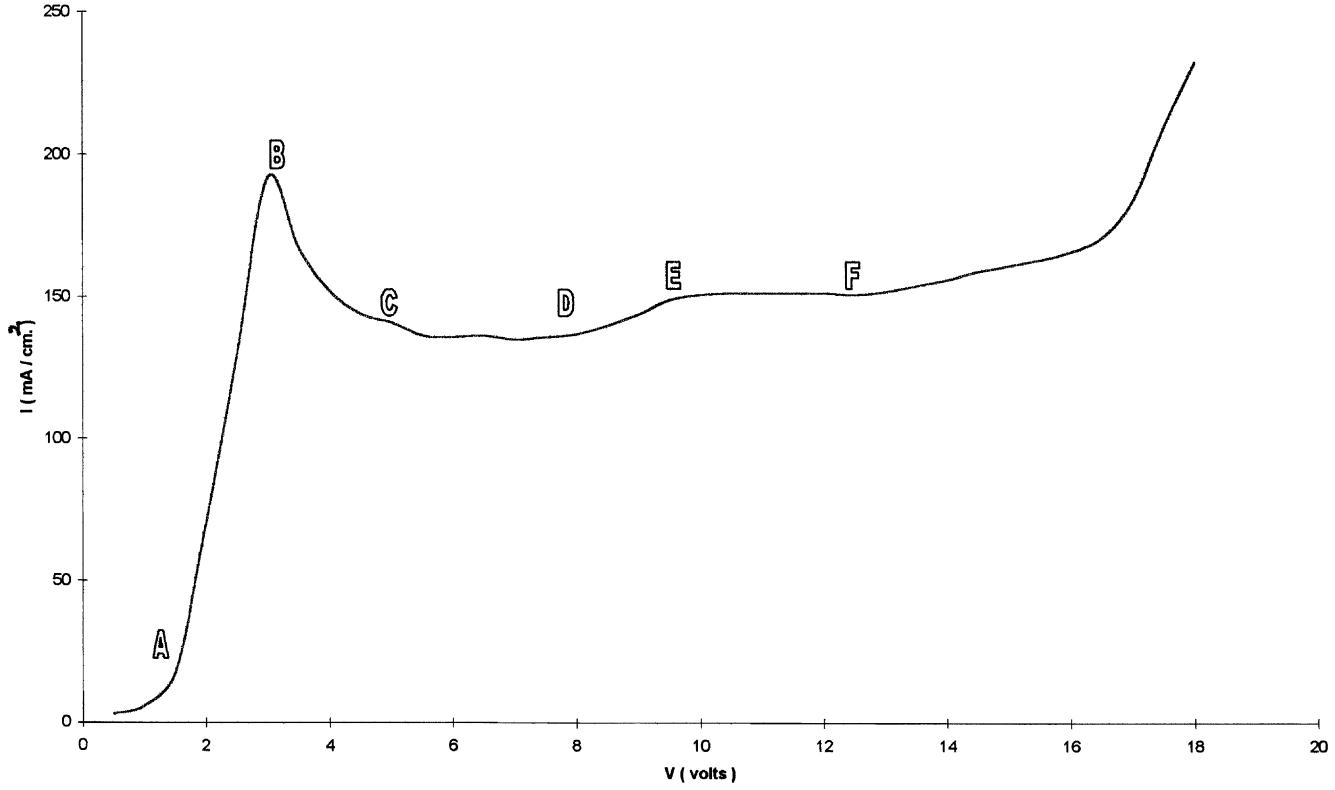


Fig. 1. Current–voltage curve showing anodic etching and electropolishing of n-type $Bi_2Se_{0.3}Te_{2.7}$: 0.2 wt.% of SbI_3.

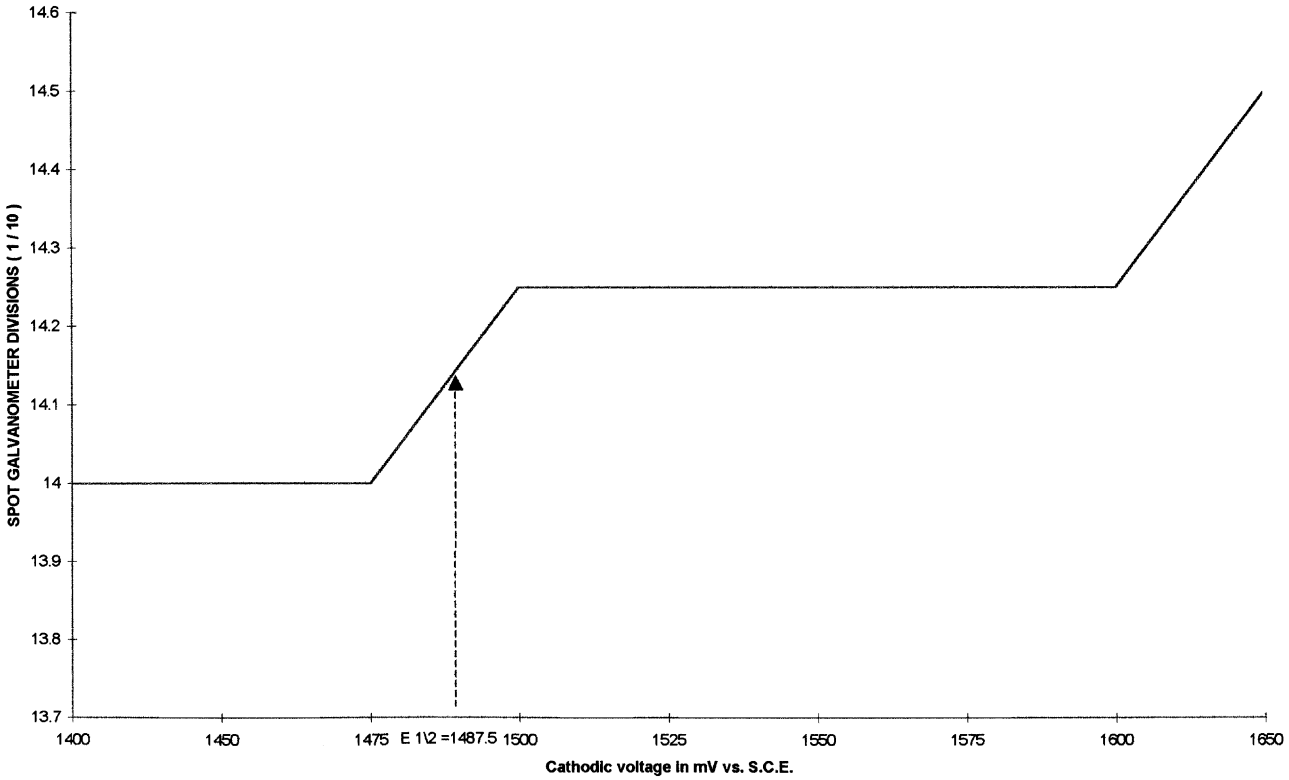


Fig. 2. Electropolishing of n-type $Bi_2Se_{0.3}Te_{2.7}$: 0.2 wt.% of SbI_3 and polarographic analysis of Se in the electropolished electrolyte: Half wave potential ($E_{1/2}$) = −1487.5 mV w.r.t. S.C.E. using DME, $E_{1/2}$ reported = −1500 mV w.r.t. S.C.E., diffusion current = 0.075 × 2.5 μA = 0.1875 μA, temperature 25°C, supporting electrolyte >0.1 M NH_4Cl, 0.003% gelatine, pH 6.8, concentration of Se in 30 cm^3 of polarographic cell determined = 0.407×10^{-3} g ions/l.

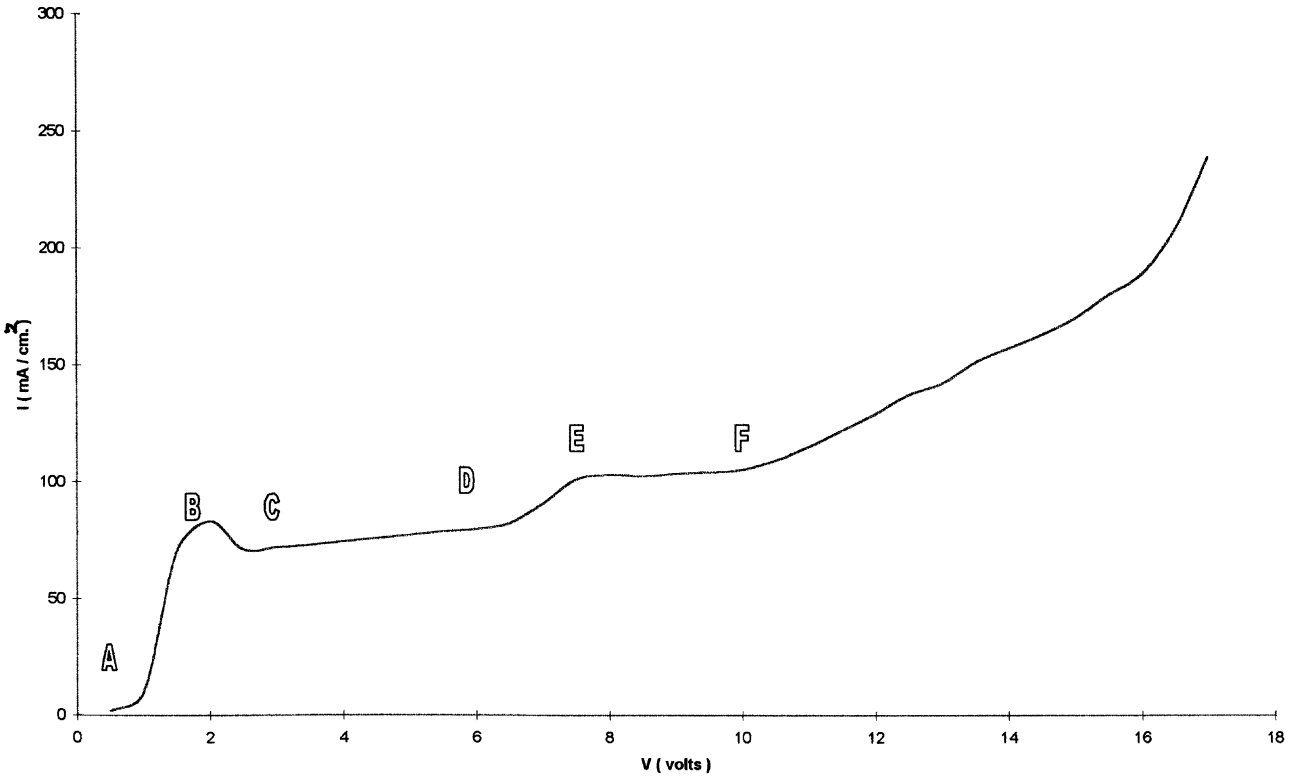


Fig. 3. *I–V* nature of the curve showing anodic etching and electropolishing of p-type $Bi_{0.5}Sb_{1.5}Te_{3.0}$.

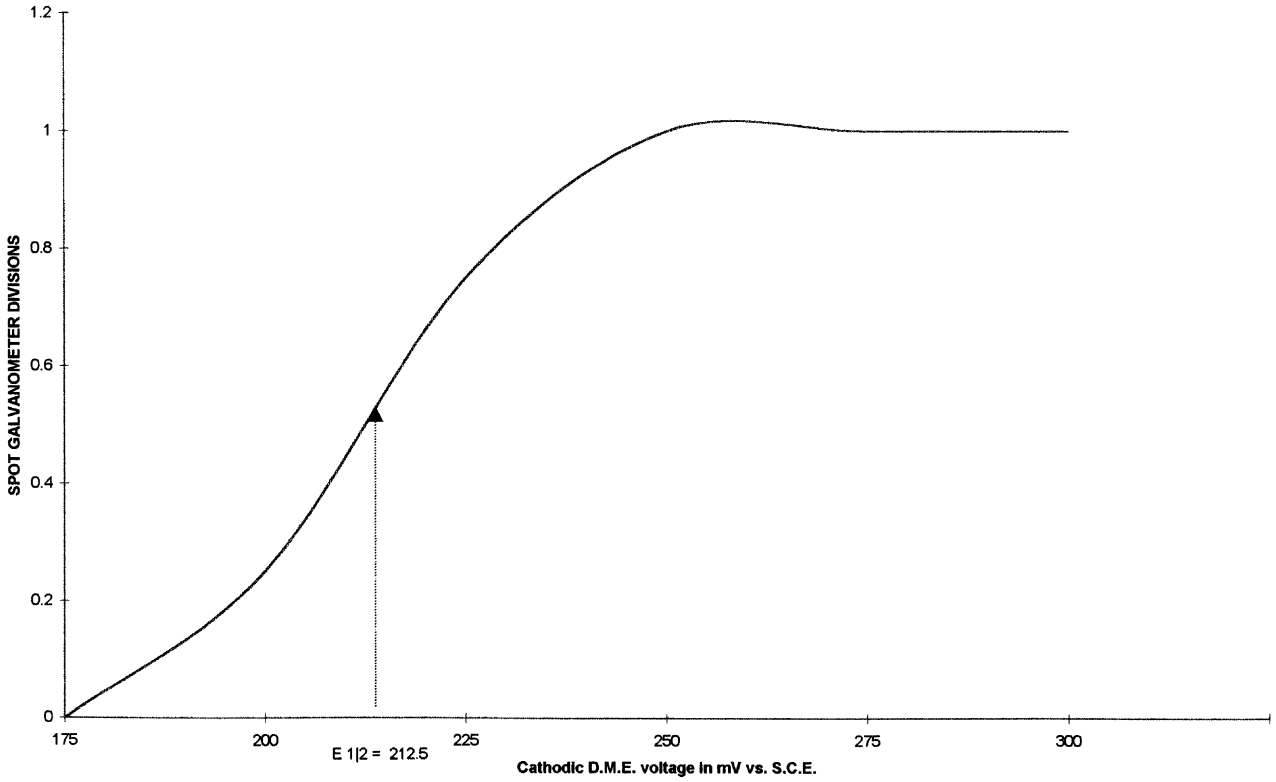


Fig. 4. Electropolishing of $Bi_{0.5}Sb_{1.5}Te_{3.0}$ and polarographic analysis of Sb in the electropolished electrolyte: $E_{1/2} = -212.5$ mV w.r.t. S.C.E., $E_{1/2}$ reported $= -180$ mV w.r.t. S.C.E., diffusion current $= 0.075$ μA, supporting electrolyte $= 0.5$ M HCl, temperature 26°C, concentration of Sb ions in 25 cm^3 of polarographic cell determined $= 1.819 \times 10^{-5}$ g ions/l.

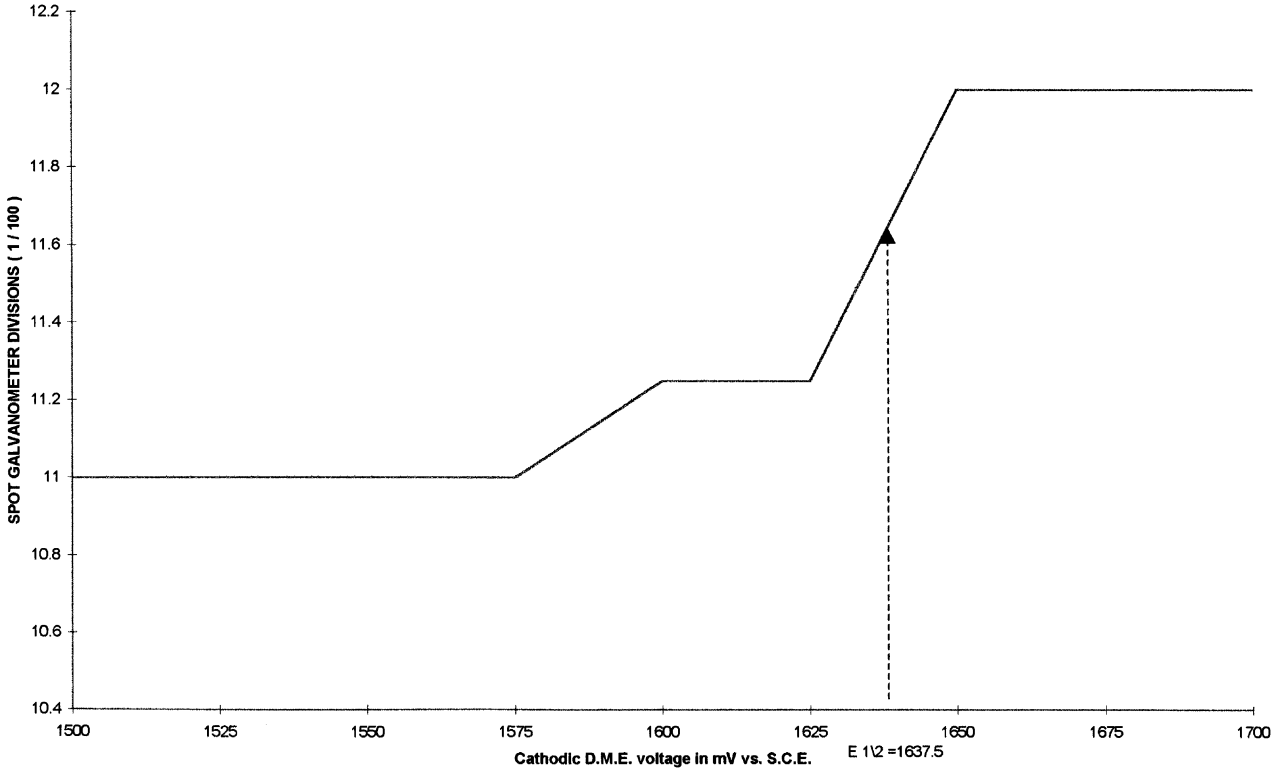


Fig. 5. Electropolishing of $Bi_2Se_{0.3}Te_{2.7}$: 0.2 wt.% of SbI_3 and polarographic analysis of Te in the electropolished electrolyte: $E_{1/2} = -1637.5$ mV w.r.t. S.C.E., $E_{1/2}$ reported $= -1660$ mV w.r.t. S.C.E., diffusion current $= 0.075 \times 75$ μA $= 5.625$ μA, supporting electrolyte $= 0.1$ M NaOH, 0.03% gelatine.

the rate of anodic etching and polishing in a regulated manner. The observed uniformity of the electropolished surfaces for both n and p-type Bi_2Te_3 based alloys is due to the coalescence of the oxide molecules with one another which increases the particle size of the oxide constituents in the polished layer. As far as the chemical composition of the etched parts of n and p-type material is concerned, these are the tartarates of bismuth and antimony in addition to K_2TeO_3 in the case of p-type material and Na_2SeO_3 and Na_2TeO_3 in the case of n-type material in electropolished electrolyte.

4. Conclusion

It is to be noted that the contents of selenium and antimony in the form of ionic concentration in the electropolished electrolyte for particular area of n and p-type Bi_2Te_3 based alloys have been determined both qualitatively and quantitatively by means of polarographic technique whereas only the qualitative part of the tellurium analysis indicating its diffusion current and half wave potential has been determined polarographically in the electropolished electrolyte and the quantitative aspect of the tellurium content by polarography is still under evaluation.

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